

Solution:

- (a) **Feasibility.** By construction, every edge inserted into A goes between v_i and some vertex in $\{v_1, \dots, v_{i-1}\}$. If we orient all chosen edges so that they go forward in the order $v_1 \prec v_2 \prec \dots \prec v_n$, then no edge ever points from a later vertex to an earlier one. Hence A admits the topological order $v_1, \dots, v_n$ and is acyclic.

Approximation ratio. Let A be the edge set returned by the algorithm, and let E^* be an optimal maximum-cardinality acyclic subgraph of G .

Observe that every edge (u, v) with $u \neq v$ appears exactly once among the union of all $\text{IN}_i \cup \text{OUT}_i$, namely when the later of u, v is being processed. Hence

$$\sum_{i=2}^n (|\text{IN}_i| + |\text{OUT}_i|) = |E|.$$

By the choice in the algorithm, the edges added at step i is

$$\max\{|\text{IN}_i|, |\text{OUT}_i|\} \geq \frac{1}{2}(|\text{IN}_i| + |\text{OUT}_i|).$$

Summing over i gives

$$|A| = \sum_{i=2}^n \max\{|\text{IN}_i|, |\text{OUT}_i|\} \geq \frac{1}{2} \sum_{i=2}^n (|\text{IN}_i| + |\text{OUT}_i|) = \frac{1}{2} |E|.$$

Therefore

$$|A| \geq \frac{1}{2} |E| \geq \frac{1}{2} |E^*|,$$

The approximation ratio is $\frac{1}{2}$.

- (b) **Feasibility.** Let π be the random order of vertices $(v_1, \dots, v_n)$. By construction, every edge in A goes forward in the random DAG order π . Hence π itself is a valid topological sort of the subgraph (V, A) , so (V, A) is a DAG.

Expected approximation ratio. Let $E^* \subseteq E$ be an optimal maximum-cardinality acyclic subgraph, so $|E^*| = \max\{|A'| : (V, A') \text{ is a DAG}\}$. Note that trivially $|E^*| \leq |E|$.

For any fixed edge $e = (u \rightarrow v) \in E$, under a uniformly random order π we have

$$\Pr[e \text{ is kept in } A] = \Pr(u \text{ precedes } v \text{ in } \pi) = \frac{1}{2}.$$

Define the indicator random variable

$$X_e = \begin{cases} 1, & \text{if } e \in A, \\ 0, & \text{otherwise.} \end{cases}$$

Then $\mathbb{E}[X_e] = \frac{1}{2}$, and linearity of expectation gives

$$\mathbb{E}[|A|] = \mathbb{E}\left[\sum_{e \in E} X_e\right] = \sum_{e \in E} \mathbb{E}[X_e] = \sum_{e \in E} \frac{1}{2} = \frac{1}{2} |E|.$$

Since $|E| \geq |E^*|$, it follows that

$$\mathbb{E}[|A|] = \frac{1}{2} |E| \geq \frac{1}{2} |E^*|.$$

The expected approximation factor is $\frac{1}{2}$.